

SANIA SHAFIQ

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ABOUT MYSELF

Ms. Sania Shafiq has completed her Master of Philosophy in Mathematics from Riphah International University, Faisalabad, Pakistan. She is deeply committed to advancing research in the field of Applied Mathematics and their applications to physical systems. Her research interests also include exploring the role of mathematics in technology, medical research, and applied sciences. Passionate about interdisciplinary collaboration, she aims to contribute meaningfully to scientific and technological innovation through mathematical research.

EDUCATION

2023 – 2025 **Master of Philosophy (MPhil) in Mathematics, Riphah International University, Faisalabad, Pakistan**
Field of study: Applied Mathematics. CGPA: 3.75/4. Supervisor: Prof. Dr. Zulfiqar Ali.
Thesis: Classical and Non-Classical Symmetry Analysis & Invariant Solutions of Some Modified Partial Differential Equations.
Conducted advanced research on classical and non-classical symmetry analysis of nonlinear PDEs, including the modified Allen–Cahn and Soliton–Burgers equations, deriving invariant solutions via Lie symmetry methods and applying the results to image scrambling, highlighting practical applications of symmetry-based analytical techniques. Major coursework: Symmetries and Exact Solutions of PDEs, Mathematical Techniques for Boundary Value Problems, Computational Mathematics, Advanced Fuzzy Linear Programming. Additional coursework: Advanced Ring Theory, Advanced Semigroup Theory, Advanced PDEs, Graph Theory, providing a strong foundation in both theoretical and applied mathematics.

2019 – 2023 **Bachelor of Science (BS) in Mathematics, Govt. College Women University, Faisalabad, Pakistan**
Field of study: Mathematics. CGPA: 3.82/4.
Major subjects: Calculus, Numerical Analysis, Partial Differential Equations, Discrete Mathematics, Statistical Mathematics, Algebra, Real and Complex Analysis, Graph Theory, Operational Research, Theory of Optimization. Supplemented with minor courses in Physics (including Mechanics) and Computer Science (such as Programming and Applied Statistics), fostering interdisciplinary analytical and computational skills.

2017 – 2019 **Intermediate (FSc, Engineering), Abdul Salam College, Faisalabad, Pakistan**
Field of study: Pre-Engineering. Grade: A.

2015 – 2017 **Matriculation (Science Group), The Punjab School System, Faisalabad, Pakistan**
Field of study: Science. Grade: A+.

ACADEMIC EXPERIENCE

15/08/2023 – Ongoing **Lecturer, Mathematics, City High School, Faisalabad, Pakistan**
Teaching mathematics courses to secondary-school students.

LANGUAGES

Urdu Native

English IELTS Academic – Overall Band 6.5 (CEFR B2). Listening 6.5, Reading 6.5, Writing 6.0, Speaking 6.0.

SCIENCE AND TECHNOLOGY SPECIALTIES

- Nonlinear partial differential equations: classical and non-classical Lie symmetry analysis, similarity reductions and invariant solutions
- Nonlinear wave phenomena and evolution equations: soliton, kink, lump and breather solutions via the Khater, Jacobi elliptic function and related analytical methods
- Dynamical systems: bifurcation, stability and parameter-sensitivity analysis of physical and biomathematical models
- Applications: mathematical modelling in materials science, biomedical sciences, wave propagation, climate modelling, image processing and quantum physics

Currently Working On

- Inverse problems and parameter estimation: Bayesian inference and symmetry-reduced formulations for nonlinear PDE models
- Scientific machine learning: Physics-Informed Neural Networks (PINNs) for forward and inverse problems
- Numerical methods: Finite Element Method (FEM) using DOLFINx (FEniCSx) and computational mathematics
- Optimization: optimization techniques, including derivative-free optimization (DFO), for parameter estimation and model calibration

RESEARCH PAPERS

Manuscripts under review:

- Shafiq, S. & Baloch, S. A. – Lie Symmetry Analysis and Lump Solutions of the (3+1)-Dimensional Modified Soliton–Burgers Equation with Applications in Nonlinear Wave Propagation.
- Shafiq, S. & Baloch, S. A. – Propagation of Dispersive Structured Soliton Dynamics in the Modified Regularized Long Waves Equation with Dispersive Effects via Advanced Computational Techniques.
- Baloch, S. A. & Shafiq, S. – Comparative Analysis of Microtubules as Electrical Conduits in Cellular and Neuronal Functions: Lump, Lump-One-Strip, Lump-Periodic, Breather, and Invariant Solutions with Applications in Bioengineering.
- Baloch, S. A., Shafiq, S., Siddique, I., El-Naggar, M. A. & Abdullaev, S. – Lie Symmetry Reductions and Computational Analysis of Exact Solutions for the (2+1)-Dimensional Generalized Benjamin–Ono Equation.

Manuscript in preparation:

- Shafiq, S. & Kungurtsev, V. – Symmetry-Reduced Bayesian Inverse Problems for Allen–Cahn Dynamics: Similarity Variables for Efficient Parameter Identification.

TECHNICAL COMPETENCIES

Programming	C++, Python, HTML, CSS, basic Julia (currently strengthening for scientific computing use)
Tools	MATHEMATICA, LaTeX and MAPLE for mathematical modelling and research; working knowledge of DOLFINx (FEniCSx); proficient in Microsoft Office Suite (Word, Excel, PowerPoint) – Microsoft Certified
Other training	Web development – Certified by Erozgar, Pakistan

HONOURS AND AWARDS

- **2025** – 2nd Position, Poster Competition, International Conference on Mathematical Sciences & Emerging Technologies (AS-ICMSET 2025), hosted by Abdus Salam School of Mathematical Sciences, GC University Lahore. Evaluated by an international panel including judges from the International Mathematical Union (Japan), European Mathematical Society (Austria), and International Mathematics Master (Netherlands).
- **2025** – Appreciation certificate for an invited talk at the 4th International Conference on Mathematical Empowerment and Future Trends (Nov 20–22, 2025), jointly organized by GCUF & GCWUF, presenting "Lie Symmetry Analysis and Lump Solutions of the (3+1)-Dimensional Modified Soliton–Burgers Equation with Applications in Nonlinear Wave Propagation."
- **2024** – Appreciation certificate for 3rd Position, Poster Competition on Mathematics for a Sustainable Future, among 6 universities of Faisalabad, Department of Mathematics, GCWUF.
- **2024** – Participation certificate, Seminar Series: Applications of Mathematics in Artificial Intelligence (3D Human Pose Estimation), Department of Mathematics, Riphah International University, Faisalabad.
- **2024** – Participation certificate, Seminar Series: Fractional Analysis of One-Dimensional Schrödinger Model using Caputo-Type Fractional Derivatives, Department of Mathematics, Riphah International University, Faisalabad.
- **2023** – Participation certificate, 2-day training workshop on Deep Learning Models for Image Recognition and Dynamical Simulation, Department of Mathematics, Govt. College University, Faisalabad.
- **2023** – Appreciation certificate for 3rd Position, Poster Competition on Applications of Mathematics, Department of Mathematics, GCWUF.

REFERENCES

Dr. Zulfiqar Ali, Professor, Head of Department
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